

# Jianjian Gao

POSTDOC RESEARCH ASSOCIATE

*School of Data Science, University of Virginia, Charlottesville, Virginia, USA*

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🔗 <https://jiaoyang2018.github.io/>

## Research Interests

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Computational social science; Science of science; Causal inference; Network science

I mainly conduct quantitative research in the field of Science of Science and Computational Social Science using various quantitative methods, including network analysis, causal inference and machine learning.

## Academic Experience

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2023- **Postdoc research associate**, School of Data Science, University of Virginia.

present Supervisor: Assistant Professor Alexander J. Gates ([hgt6rn@virginia.edu](mailto:hgt6rn@virginia.edu))

2017- **Research assistant**, Department of Energy and Power Engineering, Tsinghua University.  
2018

## Education

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### Queen Mary University of London

*London*

PHD IN BUSINESS AND MANAGEMENT (COMPUTATIONAL SOCIAL SCIENCE)

*2018 - 2022*

- Supervisor: Professor Pietro Panzarasa ([p.panzarasa@qmul.ac.uk](mailto:p.panzarasa@qmul.ac.uk))

### Institute of Engineering Thermophysics, Chinese Academy of Sciences

*Beijing*

MASTER IN POWER ENGINEERING

*2013 - 2016*

- Supervisor: Professor Hui Hong; GPA: 84.41/100

### Shandong University

*Shandong*

BACHELOR IN THERMAL ENERGY AND POWER ENGINEERING

*2009 - 2013*

- GPA: 87.91/100. Outstanding Graduates Awards of Shandong Province (2013); National Endeavour Scholarship (2012); Research and Innovation of Advanced Personal (2011); The First Prize Scholarship (2010); Merit student (2010)

## Industry Experience

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2016- **Assistant engineer**, Beijing Petrochemical Engineering Co., LTD.(BPEC)  
2017

## Publications

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## UNDER REVIEW

Alexander J. Gates, **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science and the diffusion of ideas. Nature Communications. <https://arxiv.org/abs/2404.05861>

**Jianjian Gao**, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. Recently submitted.  
[https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=6907438](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6907438)

## WORKING PAPERS

**Jianjian Gao**, Xiancheng Li. Citation Channelling: How large language models reshape scholarly knowledge search.

**Jianjian Gao**, Alexander J. Gates. The changing landscape of academic leadership: A study of presidents at US R1s.

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. Economic integration and climate change laws.

Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. The diversity of philanthropic boards and the diffusion of scientific policy.

**Jianjian Gao**, Pietro Panzarasa. Regionalization of international environmental cooperation: Evidence from community structure analysis.

## PUBLISHED

**Jianjian Gao**, Alexander J. Gates. Knowledge Sovereignty: A Path to Independence or Isolation? Research Policy. 2026, 55(8):105557 (<https://doi.org/10.1016/j.respol.2026.105557>)

Carattini Stefano, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, and Pietro Panzarasa. What does network analysis teach us about international environmental cooperation? Ecological Economics. 2023, 205:107670. **(One chapter of the PhD thesis)**

Stefano Carattini, Sam Fankhauser, **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. The global network of environmental agreements: a preliminary analysis. Annual Bank Conference on Development Economics 2019, Washington, DC.

**Jianjian Gao**. International environmental cooperation and climate change laws: A quantitative analysis. 2022. Queen Mary University of London **(PhD thesis)**.

Hui Hong, **Jianjian Gao**, Wanjun Qu, et al. Thermodynamic analyses of the solar-driven Kalina cycle having a variable concentration ratio. Applied Thermal Engineering. 2017, 126: 997-1005. **(Part of Master thesis)**

**Jianjian Gao**, Hui Hong, Jie Sun, et al. Study on the performance of concentrated solar Kalina cycle with variable mirror area parabolic trough collector. Journal of Engineering Thermophysics. 2016, 37(8):1595-1601. (EI)

Hao Zhang, Hui Hong, **Jianjian Gao**, et al. Thermodynamic performance of a mid-temperature solar fuel system for cooling, heating and power generation. Applied Thermal Engineering, 2016, 106:1268-1281.

## Patents

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Hui Hong, Jie Sun, **Jianjian Gao**. Wide-range irradiation and without energy storage concentrating solar energy Kalina power generation system and method. Beijing: CN105156285A, 2015.12.16.

## Presentations

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**NetSci**: International School & Conference on Network Science; **IC2S2**: International Conference on Computational Social Science; **ICSSI**: International Conference on the Science of Science; **Sunbelt**: Sunbelt Conference of the International Network for Social Network Analysis (INSNA); **ICA**: Annual International Communication Association Conference; **ISSI**: International Society for Scientometrics and Informetrics.

ICSSI, June 29-July 1, 2026, Denver, US – **Jianjian Gao**, Alexander J. Gates. The geography of scientific recognition across disciplines. (Talk)

ICSSI, June 29-July 1, 2026, Denver, US – **Jianjian Gao**, Alexander J. Gates. Knowledge Sovereignty: A Path to Independence or Isolation? (Poster)

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. Geographic inequality in contemporary art exhibitions arises from nationality bias and unequal returns to places. (Poster)

NetSci, June 1-5, 2026, Boston, US – **Jianjian Gao**, Alexander J. Gates. The geography of scientific recognition across disciplines. (Talk)

Sunbelt, June 23-29, 2025, Paris, France – Ty Benjamin Misiorek, **Jianjian Gao**, Alexander J. Gates. Connected boards, diverse leaders: How networks shape university presidencies. (Talk)

ISSI, June 23-27, 2025, Yerevan, Armenia. **Jianjian Gao**, Alexander J. Gates. Country self-preference and national research systems: A path to independence or isolation? (Talk)

ICA, June 12-16, 2025, Denver, Colorado, USA – Gates, Alexander J., **Jianjian Gao**, Indraneel Mane. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

ICSSI, Jul 1-3, 2024, DC, USA – **Jianjian Gao** and Gates, Alexander J. The changing landscape of academic leadership: A study of presidents at US R1s. (Poster)

ICSSI, Jul 1-3, 2024, DC, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Lightning talk)

IC2S2, Jul 18-20, 2024, Philadelphia, USA – Gates, Alexander J., Indraneel Mane, and **Jianjian Gao**. The increasing fragmentation of global science limits the diffusion of ideas. (Poster)

NetSci, Sep 17-25, 2020, Roma, Italy – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. International School and Conference on Network Science. (Poster)

IC2S2, 2020, MA USA – **Jianjian Gao**, Stefano Carattini, Sam Fankhauser, Caterina Gennaioli, Pietro Panzarasa. Structure and Evolution of the International Environmental Cooperation Network. 6th International Conference on Computational Social Science. (Talk)

IC2S2, 2020, MA USA – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Communities and rich clubs in the international environmental cooperation network. 6th International Conference on Computational Social Science. (Poster)

IC2S2, 2019, Netherlands – **Jianjian Gao**, Caterina Gennaioli, Pietro Panzarasa. Structure and evolution of the network of countries signing global environmental treaties. 5th International Conference on Computational Social Science. (Poster)

## Teaching Experience

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### Queen Mary University of London

*London*

TEACHING ASSISTANT

*Spring 2022*

- **Experiments for business and analytics (BUSM106).**
- This course deals with **advanced econometrics** and **R** programming for Master's students. I mainly taught students how to use **R** to perform all kinds of causal inference, including Difference in Difference (DID), Instrumental Variables (IV), Regression Discontinuity (RD), matching, etc.

### Queen Mary University of London

*London*

TEACHING ASSISTANT

*Fall 2022*

- **Quantitative Research Methods (BUSM014).**
- This course is designed for Master's students, covering basic elements of **statistical methods**, including descriptive statistics, probability, sampling, inference, and multivariate regression analysis. I mainly taught students how to use **Stata** to perform relevant analysis.

## Skills

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**Programming skills:** Python, Stata, R, Bash/shell scripting, HPC, SQL

**Computational skills:** Network science, Causal inference, Bayesian inference, Machine learning, Deep learning, Natural language processing.

**Certificates:** Neural Networks and Deep Learning

(<https://coursera.org/share/27bcb57c64bc13975dec99a9be379318>);

Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization

(<https://coursera.org/share/d790039cd1ce8b299448df4fe8f1fa5a>)

## References

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**Post-doctoral Research Advisor:** Assistant Professor Alexander J. Gates

University of Virginia

[agates@virginia.edu](mailto:agates@virginia.edu)

<https://alexandergates.net/>

**Doctoral Research Advisor:** Professor Pietro Panzarasa

Queen Mary University of London

[p.panzarasa@qmul.ac.uk](mailto:p.panzarasa@qmul.ac.uk)

<http://pietropanzarasa.wordpress.com/>

**Co-author:** Professor Sam Fankhauser

University of Oxford

[sam.fankhauser@smithschool.ox.ac.uk](mailto:sam.fankhauser@smithschool.ox.ac.uk)

<https://www.ox.ac.uk/news-and-events/find-an-expert/professor-sam-fankhauser>